Source-Based AI Recommendation Explanations and Consumer Decision-Making

Research Proposal

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Status: research design complete; pilot implementation pending. No human-participant data have been collected.

1. Introduction

AI recommendation systems have become a central mechanism through which e-commerce platforms shape consumer choices. Beyond ranking products, these systems increasingly use large language models to generate natural-language explanations for recommended products. Although such explanations may improve transparency, personalization, and decision support, they may also increase persuasion without necessarily improving trustworthiness or protecting consumer privacy.

Explanations differ not only in what they say but in where their supporting information comes from. A recommendation may be justified by a consumer's prior behavior, by other consumers' reviews, by the product's attributes, or by a combination of these sources. This project examines *source-based* AI recommendation explanations, defined as explanations that explicitly ground a recommendation in one or more such information sources, and treats the disclosed source as a design feature of the explanation. It asks:

> *How does the information source disclosed in an AI-generated recommendation explanation influence consumers' perceived relevance, source credibility, privacy concern, trust, and willingness to rely on the recommendation?*

The central tension is that a single disclosure can carry both a benefit and a cost. An explanation that cites a consumer's own browsing history may make a recommendation feel more relevant precisely because it reveals how much personal data the system has used. By comparing explanation sources in a controlled experiment that holds the recommendation itself constant, this study aims to identify when explanations support informed consumer choice and when they primarily increase persuasion.

2. Theoretical Background

2.1 Explanations in Recommendation Agents

Research on decision aids has long treated explanations as a means of building user understanding and acceptance. Gregor and Benbasat (1999) developed theoretical foundations for explanations from intelligent systems, proposing that users seek explanations when they perceive an anomaly, want to learn, or need specific knowledge to participate in problem solving, and that well-designed explanations can improve performance, learning, and user perceptions. In e-commerce, Xiao and Benbasat (2007) synthesized how the characteristics of product recommendation agents shape their use and users' evaluations, and Wang and Benbasat (2007) showed that different explanation facilities strengthen different trusting beliefs: "how" explanations increased beliefs in a recommendation agent's competence and benevolence, "why"

explanations its benevolence, and "trade-off" explanations its integrity. Explaining an agent's reasoning, in other words, shapes not only whether users trust it but which aspects of it they trust.

Large language models have renewed interest in this stream. Said's (2025) review shows that research on LLM-generated recommendation explanations remains in its early stages and highlights an important distinction between user-oriented justifications and faithful explanations of model reasoning. Feng, Feuerriegel, and Shrestha (2026) provide direct behavioral evidence that contextualized LLM-generated explanations based on prior user behavior can address cognitive and affective needs and increase consumption intention, but also that excessive detail may feel redundant or intrusive. On the technical side, Li et al. (2025) proposed CausalX, which integrates structural causal models with LLMs to improve causal consistency and reduce popularity bias in recommendation explanations. These studies improve or evaluate explanations as a whole; they say less about how consumers respond to the specific source of evidence an explanation invokes.

2.2 Personalization and Privacy

A second stream examines personalization itself. Awad and Krishnan (2006) documented a personalization–privacy paradox: consumers value personalized offerings, yet those who place greater value on information transparency are less willing to be profiled to receive them. More recent e-commerce studies show that personalization shapes both immediate and longer-term responses. Yin, Qiu, and Wang (2025) found that relevance, inspiration, and insight increase click-through intention, whereas perceived privacy infringement weakens these effects, and Hassan, Abdelraouf, and El-Shihy (2025) showed that personalized recommendations strengthen the relationships among trust, satisfaction, and loyalty. However, this stream mainly focuses on personalized recommendations rather than on the explanations that accompany them. An explanation that names the consumer's own behavior as its basis makes personalization visible, and may therefore activate the privacy concerns that personalization research has identified.

2.3 Explanation Sources as Signals

Signaling theory offers a way to understand why explanation sources might differ in their effects. In online markets characterized by information asymmetry, observable website features act as signals that buyers use to infer unobservable seller quality (Mavlanova, Benbunan-Fich, & Koufaris, 2012). An explanation source can be read in the same way. A reference to other consumers' reviews invokes evidence that appears external to the platform, whereas a reference to product specifications or to the consumer's own behavior invokes information the system itself controls. Because the consumer typically cannot inspect the underlying data, the disclosed source functions as a claim about evidence rather than as evidence the consumer has examined.

A parallel literature shows why such cues warrant caution. Yu, Pan, and Jang (2025) proposed a trust–risk dual-pathway framework in which AI-generated content quality increases purchase intention by increasing trust and reducing perceived risk, with ethical concerns and platform responsibility shaping these effects. Palod, Biswas, and Kambhampati (2026) further showed that LLM-generated explanations and reasoning traces can be persuasive yet uninformative, increasing user acceptance even when the underlying conclusion is incorrect. Together, these studies suggest that explanations should not be evaluated only by whether they increase acceptance, but through both benefit mechanisms, such as relevance and credibility, and risk mechanisms, such as privacy concern.

2.4 Research Gap

Prior research on explainable recommendation systems frequently conceptualizes explanations as general transparency mechanisms, often overlooking the specific information sources from which they are derived. While recent advances have enhanced the contextual relevance and technical credibility of recommendation explanations, a systematic comparison of how distinct sources of evidence, such as behavioral data, consumer reviews, and product attributes, affect consumer perceptions and reliance remains lacking. This study addresses that gap by varying only the disclosed source while holding the recommended product, its attributes, and the content of the explanation constant.

3. Conceptual Model and Hypotheses

The conceptual model proposes that explanation source shapes three proximal perceptions: perceived relevance, perceived source credibility, and situational privacy concern. These perceptions shape trust in the AI recommendation, and trust in turn shapes reliance. Different sources are expected to operate through different perceptions, so that the same overall outcome can arise from different mechanisms, or opposing mechanisms can offset each other.

Perceived relevance. Behavior-based explanations explicitly connect the recommendation to the consumer's prior activity, while product-feature-based explanations explicitly connect the product's attributes to preferences the consumer has stated. Both therefore provide a personalized rationale for why the recommended product fits the individual user.

H1a. A behavior-based recommendation explanation will lead to higher perceived relevance than a no-source control explanation.

H1b. A product-feature-based recommendation explanation will lead to higher perceived relevance than a no-source control explanation.

Privacy concern. Behavior-based explanations reveal that the system has used the consumer's prior browsing or shopping activity. The personalization that makes such an explanation feel relevant can simultaneously make the collection and use of personal behavioral data more salient, consistent with the personalization–privacy paradox.

H2. A behavior-based recommendation explanation will produce greater privacy concern than review-based, product-feature-based, and no-source explanations.

Source credibility. A review-based explanation makes an identifiable external information category, other consumers' evaluations, salient as evidence supporting the recommendation. In signaling terms, it points to evidence that appears independent of the system, giving the consumer an additional basis for evaluating the recommendation rather than requiring reliance on the AI system's assertion alone.

H3. A review-based recommendation explanation will produce greater perceived source credibility than a no-source explanation.

Trust and reliance. Relevance and credibility give consumers reasons to regard a recommendation as competent and dependable, whereas privacy concern introduces doubt about how the system treats the consumer. Trust, in turn, is expected to increase willingness to act on the recommendation.

H4. Perceived relevance (H4a) and perceived source credibility (H4b) will be positively associated with trust in the AI recommendation, whereas privacy concern (H4c) will be negatively associated with trust.

H5. Greater trust in the AI recommendation will be associated with greater willingness to rely on the recommendation.

Indirect effects. Combining these predictions, explanation source is expected to affect reliance indirectly. For behavior-based explanations in particular, the model implies two opposing indirect paths, one positive through relevance and one negative through privacy concern, so that the net effect depends on the balance between personalization benefits and privacy costs.

H6. Recommendation explanation source will influence reliance indirectly through perceived relevance, source credibility, privacy concern, and subsequent trust.

Mixed-source explanations. Combining behavioral, review, and product-feature information may strengthen relevance and credibility, but it may also increase privacy concern or informational complexity. Rather than assuming that more information is necessarily better, the study treats this condition as exploratory.

RQ1. How does a mixed-source explanation affect perceived relevance, source credibility, privacy concern, trust, and reliance relative to single-source explanations?

4. Method

4.1 Design and Context

The study uses a five-condition between-subjects online experiment. Participants imagine shopping online for wireless headphones and are told that they want good sound quality, long battery life, a comfortable fit, and reasonable value. An AI shopping assistant then recommends the same fictional product, the AeroSound X5 Wireless Headphones, in every condition. A fictional product minimizes preexisting brand attitudes while preserving a realistic consumer decision. The product, its attributes, the interface, tone, and formatting are held constant; only the information source disclosed in the explanation varies.

4.2 Stimuli

Each source-disclosing explanation follows the same two-part structure. The first sentence discloses the information source; the second names the same four attributes in every condition: *"This information highlighted sound quality, battery life, comfort, and overall value as relevant to the recommendation"* (in the mixed-source condition, "These sources highlighted…"). Holding the highlighted attributes constant ensures that conditions differ in the claimed source of evidence rather than in which product attributes are emphasized. Table 1 summarizes the source disclosures.

Table 1. Source disclosure by experimental condition

Condition	Explanation (first sentence)
C1 Control	The AI shopping assistant identified this product as a suitable option for you. (No source is disclosed and no attribute sentence follows.)
C2 Behavior-Based	…recommended this product based on your recent browsing and shopping activity.
C3 Review-Based	…recommended this product based on patterns found in customer reviews.
C4 Product-Feature-Based	…recommended this product based on how the product's features match the preferences you provided.
C5 Mixed-Source	…recommended this product based on a combination of your recent browsing and shopping activity, patterns found in customer reviews, and how the product's features match the preferences you provided.

4.3 Measures

Unless otherwise noted, constructs are measured with multi-item seven-point Likert scales (1 = strongly disagree, 7 = strongly agree) adapted from research on recommendation quality, source credibility, information privacy concern, and trust in e-commerce and recommendation agents. Items within each construct are presented in random order. Table 2 lists the measures.

Table 2. Measures

Construct	Items	Example item
Perceived relevance	3	The recommended product seems well suited to what I am looking for.
Perceived source credibility	3	The information supporting this recommendation seems credible.
Situational privacy concern	3	I feel uneasy about the possibility that my online activity was used to generate this recommendation.
Trust in the AI recommendation	4	I believe the AI shopping assistant provides reliable recommendations.
Reliance intention	3	I would rely on this recommendation when making my purchase decision.
Behavioral reliance	1 (binary)	Choose the AeroSound X5 recommended by the AI (= 1) or continue searching for another pair of headphones (= 0).

Behavioral reliance is analyzed separately from reliance intention because it captures a choice rather than a stated willingness. Five manipulation-check items, presented after all outcome measures, assess whether participants perceived the recommendation as based on their prior behavior (MC1), other consumers' reviews (MC2), product features (MC3), multiple sources (MC4), and whether the explanation clearly identified its source (MC5). Short descriptive and control measures (age, gender, online shopping frequency, experience with AI recommendation tools, and product familiarity) follow.

4.4 Procedure

After providing informed consent, participants read the shopping scenario, are randomly assigned to one condition, and view a single recommendation. They cannot return to the stimulus page after leaving it, which prevents them from re-inspecting the manipulation after seeing later questions. They then complete the relevance, credibility, privacy, trust, and reliance-intention measures, make the behavioral decision, and complete the manipulation checks and control questions. A debriefing explains that the AeroSound X5 is fictional, that the scenario was created for research purposes, and that participants received different source descriptions.

4.5 Sample Size

An a priori power analysis for a five-group one-way ANOVA ($\alpha = .05$, power = .80) with a design-stage effect-size assumption of Cohen's $f = .20$ indicates a minimum analyzable sample of approximately 305 participants. Because some responses may be excluded under data-quality criteria to be specified before collection, the recruitment target is 360 participants, or about 72 per condition. The required sample is sensitive to this assumption (approximately 536 participants for = .15 and 196 for $f = .25$), and because the hypotheses are tested primarily through two-condition contrasts, it also matters that 72 participants per condition provide 80% power to detect a standardized difference of about $d = 0.47$. Detecting $d = 0.40$ would require roughly 100 per condition. The calculation does not establish adequate power for every indirect effect, and the target will be revisited before preregistration in light of pilot evidence.

4.6 Pilot Study

The study will be piloted in two stages. In the first, a small number of internal testers will check the experiment flow, random assignment, stimulus presentation, response recording, and usability on desktop and mobile devices. In the second, an independent sample of approximately 50 to 75 participants will be used to evaluate whether the five conditions are distinguished as intended, whether the stimuli are matched in clarity and tone, whether the scales show reasonable internal consistency, whether conceptually adjacent constructs such as source credibility and trust remain empirically distinguishable, and whether the behavioral decision produces meaningful variation. The pilot is intended for design refinement rather than hypothesis testing. Depending on the results, the materials will be frozen, revised and re-piloted, or redesigned. Recruitment will proceed only after applicable research-ethics requirements have been addressed.

4.7 Ethical Considerations

The behavior-based and mixed-source conditions state that the recommendation drew on the participant's recent browsing and shopping activity, although no such data are collected. This is a mild deception, and it is necessary: telling participants in advance that the claim is hypothetical would remove the very cue under study. Its risks are limited, because no personal data are accessed, the scenario concerns an everyday purchase, and the product is fictional. The debriefing will explain that the recommendation did not use participants' actual activity, that different participants saw different source descriptions, and why the design required this, and participants who object will be able to withdraw their responses. Before recruitment, the study will be reviewed against applicable research-ethics requirements, including whether institutional review or exemption is required.

5. Analysis Plan

After data-quality checks, composite scores will be formed by averaging the items of each construct once internal consistency has been examined using Cronbach's alpha or McDonald's omega. Manipulation checks will then be evaluated; a failed manipulation would be interpreted as a stimulus-design problem rather than as evidence against the hypotheses.

H1 to H3 concern experimentally manipulated condition effects. One-way ANOVAs will evaluate overall differences, followed by planned contrasts that correspond directly to each hypothesis: behavior-based versus control (H1a) and product-feature-based versus control (H1b) on perceived relevance; behavior-based versus control, review-based, and product-feature-based on privacy concern (H2); and review-based versus control on source credibility (H3).

H4 will be tested with a multiple regression of trust on relevance, credibility, and privacy concern, with the expectation that the first two coefficients are positive and the third negative. H5 will be tested by regressing reliance intention on trust, and logistic regression will examine whether trust and condition predict the probability of accepting the recommended product. For H6, indirect effects along the hypothesized paths (for example, behavior-based → privacy concern → trust → reliance) will be estimated with bootstrap confidence intervals. Comparisons involving the mixed-source condition will be reported as exploratory.

Four principles will guide interpretation. Statistical significance will not be taken as evidence that an effect is practically large, so effect sizes and confidence intervals will be reported. A non-significant difference will not be treated as proof that two conditions are equivalent. Because the mediators are measured rather than randomized, associations among them will not be interpreted as causal effects in their own right. Finally, pilot results will inform the design but will not be used to make substantive claims about consumer behavior.

6. Expected Contributions

The project shifts attention from the general question of whether AI explanations increase trust to a more specific one: what happens when an AI system tells users where the information supporting its recommendation came from? Theoretically, it conceptualizes explanations as source-dependent information cues rather than generic transparency mechanisms, and it integrates benefit and risk pathways by examining how consumers balance the value of personalized information against concerns about privacy. Methodologically, it isolates the role of the disclosed source by holding the recommendation, emphasized attributes, and presentation constant, and it measures reliance both as stated intention and as a behavioral choice. Practically, the findings may help platforms design explanations that are useful, credible, and sensitive to consumer privacy, and may inform governance discussions about explanation provenance and responsible personalization.

The project is part of a broader agenda on how the provenance and verifiability of AI-generated information shape human judgment. A companion project, *Verification Theater*, asks whether citation-like cues create trust when the underlying evidence cannot be traced. Together, the two projects address complementary questions: how the type of source an AI system claims affects reliance, and whether users can verify that claim at all. The broader aim is to understand when explanations support not merely greater trust, but better-calibrated reliance.

7. Limitations

Several limitations should be acknowledged. The hypothetical scenario and fictional product support internal validity but limit generalization to real purchases with real consequences. A single product category of moderate involvement may not represent decisions in which stakes or expertise differ. Most mechanisms are measured through self-report, and because the mediators are not independently manipulated, the indirect-effect analyses can support but not establish causal pathways. Finally, the explanation discloses a source but participants cannot inspect it; the study therefore examines responses to claimed provenance, not to verified provenance. That distinction is the subject of the companion Verification Theater project.

8. Conclusion

AI recommendation systems increasingly explain themselves by naming their sources, and that choice is usually treated as a matter of transparency. This proposal argues that a disclosed source is also a persuasive claim, one whose effects depend on what kind of evidence it invokes and on whether the consumer can examine it. Behavioral data may make a recommendation feel more relevant precisely because it exposes the use of personal information; reviews may lend credibility by appearing independent of the platform; product attributes may offer relevance without that cost. By holding the recommendation, its emphasized attributes, and its presentation constant while varying only the disclosed source, the proposed experiment isolates these mechanisms and measures their consequences for both stated and behavioral reliance. Its results will not settle whether explanations are good or bad for consumers. They are intended to show which kinds of explanation support informed choice, which mainly increase persuasion, and how platforms might design explanations that earn trust rather than simply attract it.

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